QUESTION ANALYSIS

EEE329

Communication Engineering I

Question Analysis

Sessions 2022–23 to 2019–20 *4 papers, 107 questions, 164 formulas*

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1. At a glance:

This analysis covers **4 papers** (2022-23, 2021-22, 2020-21, 2019-20), with 107 questions and 563 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Amplitude modulation	125	22 %	29	DSB-SC (43); AM modulation index (29); VSB (20)
Angle modulation	85	15 %	16	Deviation and index (28); Armstrong modulator (19); Carson's rule (9)
Multiplexing and multiple access	82	15 %	15	TDM (50); Multiple access (16); FDM (10)
Digital modulation	75	13 %	12	DPSK (22); ASK, FSK and PSK (16); QAM (14)
Sampling and pulse modulation	74	13 %	14	PCM (32); Delta modulation (26); Sampling (12)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2022-23	2021-22	2020-21	2019-20	Term tests	All
Signals and spectrum	·	**13**	**6**	**6**	**5**	**30**
Distortionless transmission	·	**6**	·	·	**5**	11
Propagation modes	·	**7**	·	·	·	7
Transmission media	·	·	·	**6**	·	6
EM spectrum	·	·	**6**	·	·	6
Noise and channel capacity	·	**6**	**20**	**13**	**10**	**49**
Channel capacity	·	**6**	**16**	**6**	·	28
Cascaded noise	·	·	·	**7**	**10**	17
Thermal and transit-time noise	·	·	**4**	·	·	4
Amplitude modulation	·	**40**	**17**	**23**	**45**	**125**
DSB-SC	·	**14**	·	**12**	**17**	43
AM modulation index	·	**15**	**4**	·	**10**	29
VSB	·	**6**	**7**	·	**7**	20
SSB	·	·	·	**11**	**6**	17
Envelope detector	·	**5**	·	·	**5**	10
Sidebands and power	·	·	**6**	·	·	6
Angle modulation	·	**10**	**19**	**36**	**20**	**85**
Deviation and index	·	**5**	**10**	**13**	·	28
Armstrong modulator	·	·	·	**9**	**10**	19
Carson's rule	·	·	**4**	·	**5**	9
FM bandwidth	·	·	·	**8**	·	8
FM demodulation	·	·	·	**6**	·	6
FM and PM	·	·	**5**	·	·	5
Pre-emphasis	·	·	·	·	**5**	5

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Topic and subtopic	2022-23	2021-22	2020-21	2019-20	Term tests	All
NBFM	·	**5**	·	·	·	5
FM versus AM	·	·	·	·	·	0
Receivers	·	·	**7**	·	·	**7**
Superheterodyne	·	·	**7**	·	·	7
Sampling and pulse modulation	·	**26**	**22**	**19**	**7**	**74**
PCM	·	**15**	**5**	**5**	**7**	32
Delta modulation	·	**6**	**10**	**10**	·	26
Sampling	·	**5**	**7**	·	·	12
Quantization	·	·	·	**4**	·	4
Digital modulation	·	**24**	**19**	**28**	**4**	**75**
DPSK	·	**6**	**4**	**8**	**4**	22
ASK, FSK and PSK	·	**11**	·	**5**	·	16
QAM	·	**7**	**7**	·	·	14
QPSK	·	·	·	**8**	·	8
FSK	·	·	**8**	·	·	8
Line coding	·	·	·	**7**	·	7
Multiplexing and multiple access	·	**18**	**28**	**17**	**19**	**82**
TDM	·	**12**	**7**	**12**	**19**	50
Multiple access	·	**6**	**10**	·	·	16
FDM	·	·	**5**	**5**	·	10
Spread spectrum	·	·	**6**	·	·	6
Information theory	·	**6**	**5**	**8**	**10**	**29**
Entropy	·	**6**	**5**	**8**	**10**	29
System design	·	**7**	·	·	·	**7**
Link design	·	**7**	·	·	·	7

3. Priority list:

Must know: Amplitude modulation (125), Angle modulation (85), Multiplexing and multiple access (82), Digital modulation (75)

Likely: Sampling and pulse modulation (74), Noise and channel capacity (49), Signals and spectrum (30), Information theory (29)

Less often: Receivers (7), System design (7)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

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4. Topic pages:

4.1. Signals and spectrum:

30 marks, 5 questions, in 3 of 4 papers. Book: Lathi Ch. 1, 3; Forouzan Ch. 3.

Most often theory (3 of 5). Subtopics: Transmission media; EM spectrum; Distortionless transmission; Propagation modes.

Formulas:

Subtopic	Formula	Equation	Used in
Distortionless transmission (11)	**Distortionless channel**	$H(f) = k\, e^{-j2\pi f t_d}$	21-22 1(d), 20-21 7(b)
	Distortionless output	$y(t) = k\, x(t - t_d)$	21-22 1(d)
	Group delay	$t_g = -\frac{1}{2\pi}\frac{d\theta_h}{df} = t_d$	21-22 1(d)
Propagation modes (7)	Radio horizon (LOS)	$d \approx 4.12\left(\sqrt{h_t} + \sqrt{h_r}\right)$ km	21-22 1(b)

Symbols:

k = constant gain ↩ θ_h = phase response (rad) ↩ $\frac{4}{3}$-Earth radius ↩

t_d = constant delay (s) ↩ h_t, h_r = antenna heights (m),

Questions:

No.	Paper	Marks	Type	Question
Transmission media				
1	19-20 1(a)	6	theory	Write short notes on the following communication channels: (i) Twisted pair cable (ii) Radio wave
EM spectrum				
2	20-21 1(b)	6	theory	Briefly explain the three major bands of the electromagnetic (EM) spectrum-radio waves, microwaves and infra-red waves-in terms of their bandwidth, propagation mode and applications.
Distortionless transmission				
3	21-22 1(d)	6	derivation repeats 20-21 7(b)	Explain the conditions required for distortionless transmission.
4	20-21 TT1 7(b)	5	derivation	Explain distortion as a transmission impairment in communication systems. What are the conditions required for distortionless transmission?
Propagation modes				
5	21-22 1(b)	7	theory	Discuss the different modes of propagation of EM wave.

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4.2. Noise and channel capacity:

49 marks, 9 questions, in 3 of 4 papers. Book: Lathi Ch. 1, 13; Forouzan Ch. 3.

Most often numerical (5 of 9). Subtopics: Channel capacity; Cascaded noise; Thermal and transit-time noise.

Formulas:

Subtopic	Formula	Equation	Used in
Channel capacity (28)	**Shannon capacity**	$C = B\log_2\left(1+\frac{S}{N}\right)$ bit/s	21-22 1(c), 19-20 1(c)
	SNR for a rate	$\frac{S}{N} = 2^{C/B} - 1$	19-20 1(c)
	dB to ratio	$\text{SNR} = 10^{\text{SNR}_{dB}/10}$	19-20 1(c)
	Equally likely levels	$H = \log_2 L$	20-21 3(d)
	Information rate	$R = f_s H$ bit/s	20-21 3(d)
	Shannon capacity	$C = B\log_2\left(1+\frac{S}{N}\right)$	20-21 3(d)
	Decibels to ratio	$\frac{S}{N} = 10^{\text{dB}/10}$	20-21 5(b)
	Minimum SNR	$\text{SNR}_{min} = 2^{R/B} - 1$	20-21 5(b)
	Bandwidth for a rate	$B = \frac{R}{\log_2(1+S/N)}$	20-21 3(d)
	Attenuation	$20\log_{10}\frac{V_t}{V_r}$ dB	20-21 7(c)
	SNR from amplitudes	$\text{SNR} = (V_s/V_n)^2$	20-21 7(c)
	Signal for a rate	$V_s = V_n\sqrt{2^{R/B}-1}$	20-21 7(c)
	Attenuation	$10\log_{10}\frac{P_t}{P_r} = 20\log_{10}\frac{V_t}{V_r}$ dB	21-22 1(c)
	Signal-to-noise ratio	$\frac{S}{N} = \left(\frac{V_s}{V_n}\right)^2$	21-22 1(c)
	SNR for a rate	$\frac{S}{N} = 2^{R/B} - 1$	21-22 1(c)
Cascaded noise (17)	Thermal noise	$\overline{v_n^2} = 4kTBR$	19-20 1(d)
	Cascade noise resistance	$R_{eq} = R_1 + \frac{R_2}{A_1^2} + \frac{R_3}{A_1^2 A_2^2} + \cdots$	21-22 7(b), 20-21 7(d), 19-20 1(d)
Thermal and transit-time noise (4)	**Thermal noise power**	$P_n = kTB$	20-21 1(c)
	Thermal noise voltage	$V_n = \sqrt{4kTBR}$	20-21 1(c)

Symbols:

C = largest error-free rate (bit/s) ↩

B = channel bandwidth (Hz) ↩

L = number of levels ↩

f_s = symbols per second ↩

S/N = plain power ratio ↩

V_t, V_r = sent and received amplitude (V) ↩

R = wanted data rate (bit/s); resistance (Ω) ↩

P_t, P_r = sent and received power (W) ↩

V_s, V_n = signal and noise amplitude (V), same resistance ↩

T = temperature (K)

R_i = noise resistance at stage i's input

A_i = voltage gain of stage i

Questions:

No.	Paper	Marks	Type	Question
Channel capacity				
1	21-22 1(c)	6	numerical repeats 20-21 7(c)	We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10 V is transmitted, and the receiver detects a signal with an amplitude of 2 V and noise with an amplitude of 5 mV. (i) Calculate the attenuation. (ii) Determine the maximum channel capacity. (iii) To achieve a data rate of 100 Mbit s^{-1}, find the required signal level.
2	20-21 3(d)	10	numerical	An analog signal having 4 kHz bandwidth is sampled at 1.25 times the Nyquist rate, and each sample is quantized into one of 256 equally likely level. Assuming the samples to be statistically independent: (i) What is the information rate of the source? (ii) Can the output of this source be transmitted without error over the AWGN channel with a bandwidth of 10 kHz and S/N ratio of 20 dB? (iii) Find S/N ratio required for error-free transmission of part ii. (iv) Find the bandwidth required for an AWGN channel for an error free transmission of output of this source if S/N ratio is 20 dB
3	20-21 5(b)	6	numerical	A voice grade telephone channel has a bandwidth of 3400 Hz. If the signal to noise ratio (SNR) on the channel is 30 dB, determine the capacity of the channel. If the above channel is to be used to transmit 4.8 kbps of data, determine the minimum SNR required on the channel.
4	20-21 TT1 7(c)	0	numerical	We have a communication channel with a bandwidth of 5 MHz. A signal with an amplitude of 10V is transmitted, and the receiver detects a signal with an amplitude of 2V and noise with an amplitude of 5 mV.
5	19-20 1(c)	6	numerical	States Shannon's capacity theorem. A user wishes to transmit data over a channel having bandwidth B = 3000 Hz. If the SNR is 9.08 find the theoretical maximum bit rate of the channel. Also find the required SNR if the desired bitrate of the channel is doubled.
Cascaded noise				
6	21-22 TT1 7(b)	5	derivation repeats 19-20 1(d)	Find the equivalent noise resistance for several amplifiers in cascade.
7	20-21 TT1 7(d)	5	derivation repeats 19-20 1(d)	Find the equivalent noise resistance for several amplifiers in cascade.
8	19-20 1(d)	7	derivation	Find the equivalent noise resistance for several amplifiers in cascade.
Thermal and transit-time noise				
9	20-21 1(c)	4	theory	Write short note on- (i) Thermal Noise (ii) Transit Time Noise

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4.3. Amplitude modulation:

125 marks, 29 questions, in 4 of 4 papers. Book: Lathi Ch. 4.

Most often theory (11 of 29). Subtopics: DSB-SC; SSB; AM modulation index; Sidebands and power; VSB; Envelope detector.

Formulas:

Subtopic	Formula	Equation	Used in
DSB-SC (43)	Product to sum	$\cos A \cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$	19-20 1(b)
	DSB-SC of a tone	$a\cos\omega_m t\cos\omega_c t = \frac{a}{2}\cos(\omega_c-\omega_m)t + \frac{a}{2}\cos(\omega_c+\omega_m)t$	19-20 1(b)
	Square of a sine	$\sin^2 x = \frac{1}{2}(1-\cos 2x)$	19-20 1(b)
	rad/s to Hz	$f = \omega/2\pi$	19-20 1(b)
	Coherent detector output	$e_o(t) = \frac{1}{2}m(t)\cos(\Delta\omega t + \delta)$	19-20 2(c)
	DSB-SC spectrum	$S(f) = \frac{A_c}{2}[M(f-f_c)+M(f+f_c)]$	20-21 5(d)
	Coherent detector product	$V(f) = \frac{A_c}{2}M(f) + \frac{A_c}{4}[M(f-2f_c)+M(f+2f_c)]$	20-21 5(d)
	Lowest carrier	$f_c \geq W$	20-21 5(d)
	AM spectrum	$\pi A[\delta(\omega-\omega_c)+\delta(\omega+\omega_c)] + \frac{1}{2}[X(\omega-\omega_c)+X(\omega+\omega_c)]$	20-21 8(a)
	Coherent detector with errors	$e_o = [A+m(t)]\cos(\Delta\omega t+\delta)$	20-21 8(b)
	Coherent detection with errors	$e_o(t) = m(t)\cos(\Delta\omega\, t+\theta)$	21-22 2(b)
	Costas I and Q outputs	$\frac{1}{2}m(t)\cos\theta,\ \frac{1}{2}m(t)\sin\theta$	21-22 2(c)
	Costas control signal	$\frac{1}{8}m^2(t)\sin 2\theta$	21-22 2(c), 21-22 8(a)
	Square wave (±1)	$w(t) = \frac{4}{\pi}\left[\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \cdots\right]$	22-23 7(c)
	Ring modulator output	$\frac{4}{\pi}m(t)\cos\omega_c t$ after the BPF	22-23 7(c)
	DSB-SC with LO errors	$e_o = m(t)\cos(\Delta\omega t+\delta)$	22-23 7(e)
	SSB with LO errors	$e_o = m\cos(\Delta\omega t+\delta) - m_h\sin(\Delta\omega t+\delta)$	22-23 7(e)
SSB (17)	Sideband gap for filtering	$\text{gap} = 2f_L$	19-20 2(d)
	SSB (phase shift)	$\varphi_{SSB}(t) = m(t)\cos\omega_c t \mp m_h(t)\sin\omega_c t$	19-20 3(a)
	Hilbert transformer	$H(f) = -j\,\text{sgn}(f)$	19-20 3(a)
	SSB + carrier envelope	$\sqrt{(A+m)^2+m_h^2} \approx A+m$	21-22 8(b)
	Bandwidth	AM, DSB-SC: $2B$; SSB: B	22-23 7(d)

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Subtopic	Formula	Equation	Used in
AM modulation index (29)	AM wave	$s(t) = [E_c + E_m \sin \omega_m t] \sin \omega_c t$	20-21 1(d)
	Modulation index	$\mu = \dfrac{E_m}{E_c}$	20-21 1(d)
	Side frequencies	$f_c \pm f_m$, amplitude $\dfrac{\mu E_c}{2}$	20-21 1(d)
	AM antenna current	$I_t = I_c\sqrt{1 + \mu^2/2}$	20-21 8(c)
	AM wave	$\varphi_{AM}(t) = [A + m(t)] \cos \omega_c t$	21-22 3(d)
	Modulation index	$\mu = \dfrac{m_p}{A}$	21-22 3(d)
	AM antenna current	$I_t = I_c\sqrt{1 + \dfrac{\mu^2}{2}}$	21-22 2(a)
Sidebands and power (6)	Carrier power	$P_c = \dfrac{E_c^2}{2R}$	22-23 7(b), 20-21 1(e)
	Total power	$P_t = P_c\left(1 + \dfrac{\mu^2}{2}\right)$	22-23 7(b), 21-22 2(a), 20-21 1(e)
	Modulation index	$\mu = E_m/E_c$	22-23 7(b)
	Side frequencies	$f_c \pm f_m$, amplitude $\mu E_c/2$	22-23 7(b)
	Efficiency	$\eta = \dfrac{\mu^2}{2 + \mu^2}$	22-23 7(b)
VSB (20)	**VSB receiver filter**	$H_o(f) = \dfrac{1}{H_i(f + f_c) + H_i(f - f_c)}$	21-22 2(d), 20-21 4(b)
	VSB signal	$\Phi_{VSB}(f) = [M(f + f_c) + M(f - f_c)]\, H_i(f)$	21-22 2(d)
	VSB filter condition	$H_i(f + f_c) + H_i(f - f_c) = \text{constant},\ \lvert f\rvert \le B$	21-22 2(d)
	VSB two tones	$\frac{1}{2}[a + (1 - a)] \cos \omega_1 t + \frac{1}{2} \cos \omega_2 t = \frac{1}{2} m(t)$	21-22 8(c)
	VSB filter condition	$H_i(f_c + f) + H_i(f_c - f) = 1$	22-23 7(g)
	VSB bandwidth	$B + f_v$	22-23 7(g)
Envelope detector (10)	Envelope detector RC	$\dfrac{1}{\omega_c} \ll RC < \dfrac{1}{2\pi B}$	21-22 3(b)
	Envelope detector RC	$\dfrac{1}{\omega_c} \ll RC \le \dfrac{\sqrt{1 - \mu^2}}{\mu \omega_m}$	21-22 7(d)

Symbols:

$\Delta\omega$ = local-carrier frequency error (rad/s)

δ = phase error

$M(f)$ = message spectrum ↩

W = message bandwidth ↩

$\Delta\omega$, δ = frequency and phase error ↩

θ = phase error (rad) ↩

f_L = lowest message frequency ↩

m_h = Hilbert transform ↩

E_c, E_m = peak carrier and message (V)

A = carrier amplitude ↩

ω_c = carrier frequency (rad/s) ↩

I_t = rms current with modulation (A) ↩

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I_c = rms current of the carrier alone (A) ↩

R = load (Ω) ↩

P_t = total power ↩

P_c = carrier power (W) ↩

E_m, E_c = message and carrier peaks ↩

H_i = transmit filter ↩

$H_o(f)$ = receiver LPF, for $|f| \leq B$ ↩

$H_i(f)$ = VSB shaping filter ↩

a = filter gain at $f_c + f_1$ ↩

f_v = vestige width ↩

RC = load time constant (s) ↩

Questions:

No.	Paper	Marks	Type	Question
DSB-SC				
1	22-23 TT2 7(c)	0	theory	Explain switching modulators, specifically the ring modulator: circuit, working, and the case $m(t) = 0$ where the carrier is suppressed.
2	22-23 TT2 7(e)	0	derivation	Explain the effect of local oscillator phase error and frequency error in coherent detection of DSB-SC/SSB.
3	21-22 2(b)	6	theory repeats 19-20 2(c)	Explain how phase and frequency error affect the coherent detection of AM.
4	21-22 2(c)	8	theory	With appropriate block diagram, briefly explain the working principle of Costas DSB-SC receiver.
5	21-22 TT2 8(a)	7	theory	What is quadrature null effect? Utilizing proper block diagram, explain principle of Costas receiver and how it prevents quadrature null effect.
6	20-21 5(d)	0	theory	Consider a message signal m(t) with the spectrum shown in Fig. 1. The message bandwidth W = 1 kHz. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the DSBSC modulated wave s(t). This modulated wave is next applied to a coherent detector. Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: (i) the carrier frequency $f_c = 1.25$ kHz and (ii) the carrier frequency $f_c = 0.75$ kHz.
7	20-21 TT2 8(a)	5	diagram	Amplitude spectrum of a message signal shown in figure-1. Draw the spectrum of its DSB-WC and SSB-SC modulated signal.
8	20-21 TT2 8(b)	5	derivation repeats 19-20 2(c)	Explain how phase and frequency error affect the coherent detection of AM.
9	19-20 1(b)	6	numerical	For carrier frequency 100 kHz, find the frequency, sketch the spectrum, indicate the upper sideband, lower sideband and their amplitudes of the following signals. (i) $m(t) = 1 + \cos(377t)$ (ii) $m(t) = 2\sin^2(100\pi t)$ (iii) $m(t) = 3\cos(1000t)\cos(3000t)$
10	19-20 2(c)	6	derivation	Explain the effect of phase and frequency error during coherent detection of DSB-SC.
SSB				
11	22-23 TT2 7(d)	0	theory	Compare AM (DSB-WC), DSB-SC and SSB-SC in modulation, demodulation, power efficiency and bandwidth.
12	22-23 TT2 7(f)	0	theory	Disadvantages of Weaver's method of SSB generation.
13	21-22 TT2 8(b)	6	derivation	Explain how SSB can be detected using envelope detection method.
14	19-20 2(d)	3	theory	Describe the condition on message signal spectrum for SSB generation using selective filtering.
15	19-20 3(a)	8	derivation	Using proper block diagram explain phase shifting method of SSB generation.
AM modulation index				
16	21-22 2(a)	5	numerical repeats 20-21 8(c)	The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8

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No.	Paper	Marks	Type	Question
17	21-22 3(d)	10	diagram	For the AM signal with $m(t)$ in Fig. 1 given below, sketch $\Phi_A M(t)$ for $\mu = 0.5, \mu = 1$, and $\mu = 1.5$.
18	21-22 TT1 7(c)	5	numerical repeats 20-21 8(c)	The antenna current of an AM transmitter is 8 A when only the carrier is sent. But it increases to 8.93 A when the carrier is modulated by a single sine wave. Find the percentage modulation. Determine the antenna current when the percentage of modulation changes to 0.8.
19	20-21 1(d)	4	numerical	A carrier wave of frequency 10 MHz and peak value of 10 V is amplitude modulated by a 5 kHz sine wave of amplitude 6 V. Determine the modulation index, side frequencies and amplitude of side frequencies.
20	20-21 TT2 8(c)	5	numerical	The antenna current of an AM transmitter is 8 Amps, when only the carrier is sent, but it increases to 8.93A, when the carrier is modulated by a single sine wave. Find percentage modulation. Determine the antenna current when the percentage modulation changes to 0.8
Sidebands and power				
21	22-23 TT2 7(b)	0	numerical repeats 20-21 1(e)	An audio frequency signal $10\sin 2\pi 500t$ is used to amplitude modulate a carrier of $50\sin 2\pi 10^5 t$. Calculate: (i) Modulation index (ii) Sideband frequencies (iii) Amplitude of each sideband frequency (iv) Bandwidth required (v) Total power delivered to the load of 600 Ω (vi) Transmission efficiency
22	20-21 1(e)	6	numerical	An audio frequency signal $10\sin(2\pi 500t)$ is used to amplitude modulated a carrier of $50\sin(2\pi 105t)$ *[sic: should be* $2\pi 10^5 t$*]*. Calculate: (i) Modulation index (ii) Side-band frequencies (iii) Total power delivered to the load of 600 Ω
VSB				
23	22-23 TT2 7(g)	0	theory	What is VSB (vestigial sideband)? Why is it used?
24	22-23 TT2 7(h)	0	theory	Why is VSB-WC chosen for TV broadcasting?
25	21-22 2(d)	6	derivation	Find the condition of sideband shaping filter for VSB modulation.
26	21-22 TT2 8(c)	7	derivation	Using a message signal consists of two sinusoidal frequency components, describe the principle of VSB modulation and demodulation.
27	20-21 4(b)	7	derivation	Explain how the message signal is reconstructed from VSB signal.
Envelope detector				
28	21-22 3(b)	5	theory	Write a short note on the linear diode detection (envelope detector) method with proper diagrams and leveling.
29	21-22 TT1 7(d)	5	derivation	Explain the principle of envelope detector and hence find the RC requirement of envelope detector.

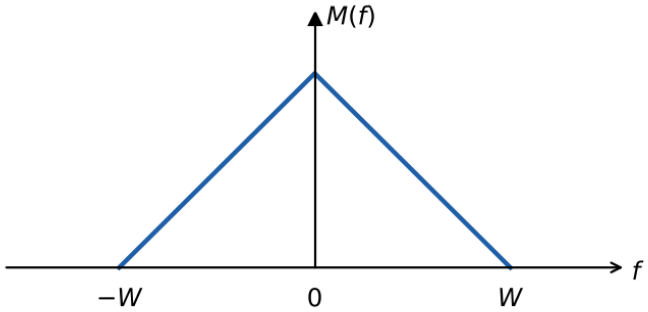


Figure 1: 20-21 5(d): Fig. 1 of the paper (redrawn)

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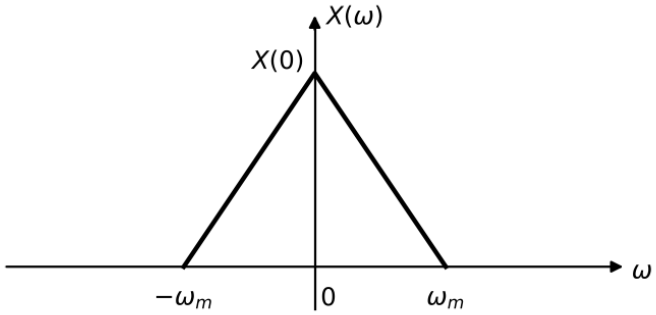


Figure 2: 20-21 TT2 8(a): Figure-1 of the paper (redrawn)

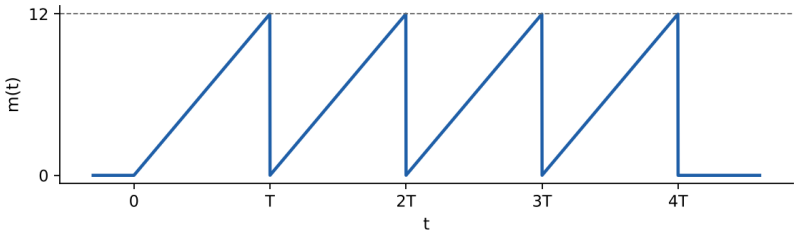


Figure 3: 21-22 3(d): Fig. 1 of the paper (redrawn): the message m(t)

4.4. Angle modulation:

85 marks, 16 questions, in 4 of 4 papers. Book: Lathi Ch. 5.

Most often theory (7 of 16). Subtopics: Deviation and index; FM bandwidth; Armstrong modulator; FM demodulation; FM and PM; Carson's rule; Pre-emphasis; NBFM; FM versus AM.

Formulas:

Subtopic	Formula	Equation	Used in
Deviation and index (28)	Instantaneous frequency	$\omega_i = \frac{d\theta}{dt},\ f_i = f_c + \frac{1}{2\pi}\frac{d\phi}{dt}$	19-20 2(b)
	Power of angle modulation	$P = \frac{A^2}{2}$	19-20 2(b)
	FM deviation	$\Delta f = k_f V_m$	20-21 2(d), 19-20 4(a)
	Modulation index	$\beta = \frac{\Delta f}{f_m}$ (or $\Delta f/B$)	19-20 4(a)
	Carson's rule	$B_T = 2(\Delta f + B) = 2B(\beta + 1)$	19-20 2(b)
	FM modulation index	$\beta = \frac{\Delta f}{f_m}$	21-22 3(c), 20-21 2(d)
	Frequency limits	$f_{max,min} = f_c \pm \Delta f$	21-22 3(c)
	Carrier swing	swing $= 2\Delta f$	21-22 3(c)
	Carson's rule	$B_T \approx 2(\Delta f + f_m)$	21-22 3(c)
FM bandwidth (8)	FM line spectrum	$A\sum_n J_n(\beta)\cos(\omega_c + n\omega_m)t$	19-20 3(b)
Armstrong modulator (19)	Multiplier	$f_c \to nf_c,\ \Delta f \to n\Delta f$	22-23 8(a), 19-20 3(c)

Subtopic	Formula	Equation	Used in
	Mixer	$f_c \to f_c \pm f_{LO}$, Δf unchanged	22-23 8(a), 19-20 3(c)
	Armstrong design	$n_1 n_2 = \dfrac{\Delta f}{\Delta f_1}$, $f_{LO} = \left\lvert \dfrac{f_c}{n_2} - n_1 f_{c1} \right\rvert$	22-23 8(a), 19-20 3(c)
FM demodulation (6)	Slope (differentiator) detector	$\dot{\varphi}_{FM} = -A[\omega_c + k_f m(t)]\sin(\cdot)$	19-20 4(b)
FM and PM (5)	Instantaneous frequency	$\omega_i = \dfrac{d\theta}{dt}$	20-21 2(c)
	PM	$\varphi_{PM} = A\cos[\omega_c t + k_p m(t)]$	20-21 2(c)
	FM	$\varphi_{FM} = A\cos\left[\omega_c t + k_f \int m\, d\alpha\right]$	20-21 2(c)
Carson's rule (9)	**Carson's rule**	$B_T \approx 2(\Delta f + f_m) = 2(\beta + 1) f_m$	20-21 4(d)
	Deviation from sensitivity	$\Delta f = k_f A_m$	20-21 8(e)
Pre-emphasis (5)	De-emphasis	$H_d(f) = \dfrac{1}{1 + jf/f_1}$, $f_1 = \dfrac{1}{2\pi RC}$	20-21 8(d)
NBFM (5)	FM wave	$\varphi_{FM}(t) = A\cos[\omega_c t + k_f a(t)]$, $a(t) = \int m(\tau)\, d\tau$	21-22 3(a)
	NBFM	$\varphi_{NBFM}(t) \approx A\cos\omega_c t - A\, k_f\, a(t)\sin\omega_c t$	21-22 3(a)
	FM signal	$A\cos[\omega_c t + k_f a(t)]$, $a(t) = \int m\, dt$	22-23 7(i)
	NBFM	$\varphi_{NBFM} \approx A\cos\omega_c t - A k_f a(t)\sin\omega_c t$	22-23 7(i)
	NBFM, tone	$A\cos\omega_c t + \frac{A\beta}{2}\cos(\omega_c + \omega_m)t - \frac{A\beta}{2}\cos(\omega_c - \omega_m)t$	22-23 7(i)
FM versus AM (0)	Carson's rule	$B_T = 2(\Delta f + B)$	22-23 7(a)

Symbols:

$\theta(t)$ = carrier angle

k_f = Hz per volt; frequency sensitivity (rad/s per volt); Hz/V ↩

B = highest message frequency ↩ ↩

f_m = modulating frequency (Hz) ↩

Δf = peak frequency deviation (Hz)

B_T = FM bandwidth (Hz) ↩

k_p = phase sensitivity ↩

Questions:

No.	Paper	Marks	Type	Question
Deviation and index				
1	21-22 3(c)	5	numerical	A 93.2 MHz carrier is frequency modulated by a 5 kHz sine wave. The resultant FM signal has a frequency deviation of 40 kHz. (i) Find the carrier swing of the FM signal. (ii) What are the highest and lowest frequencies attained by the frequency modulated signal? (iii) Calculate the modulation index for the wave.
2	20-21 2(d)	10	numerical	In an FM system, when the audio frequency is 500 Hz and modulating voltage 2.5 V, the deviation produced is 5 kHz. If the modulating voltage is now increased to 7.5 V, calculate the new value of frequency deviation produced. If the AF voltage is raised to 10 V while the modulating frequency dropped to 250 Hz, what is the frequency deviation? Calculate the modulation index in each case.

No.	Paper	Marks	Type	Question
3	19-20 2(b)	8	numerical	An angle modulated signal with carrier frequency 10 MHz is described by the equation $s(t) = 12\cos(\omega_c t + \sin 3000\pi t + 10\sin 2000\pi t)$ (i) Find the power of the modulated signal (ii) Find the frequency deviation Δf (iii) Find the β (iv) Find the bandwidth.
4	19-20 4(a)	5	numerical	In an FM system the frequency deviation is 5 kHz, when its audio modulating frequency is 400 Hz, and the audio modulating voltage is 2 V. Compute the modulation index. Also compute the frequency deviation and the modulation index if (i) if voltage is increased to 6V keeping modulation frequency unaltered, (ii) if voltage is increased to 8 V while the modulating frequency is reduced to 200 Hz.
FM bandwidth				
5	19-20 3(b)	8	theory	Explain the effect of variation of amplitude and frequency of message signal on the spectrum and bandwidth of FM.
Armstrong modulator				
6	22-23 TT3 8(a)	10	design repeats 19-20 3(c)	Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3MHz and $\Delta f = 20.48kHz$. A NBFM generator of $f_{c1} = 20kHz$ and $\Delta f = 5Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 50-kHz *[sic: should be "500 kHz"]* is readily available for frequency mixing.
7	19-20 3(c)	9	design	Design an Armstrong indirect FM modulator to generate an FM signal with carrier frequency 97.3 MHz and $\Delta f = 10.24\ kHz$. A NBFM generator of $f_{c1} = 20\ kHz$ and $\Delta f = 5\ Hz$ is available. Only frequency doublers can be used as multipliers. Additionally, a local oscillator (LO) with adjustable frequency between 400 and 500 kHz is readily available for frequency mixing.
FM demodulation				
8	19-20 4(b)	6	theory	Explain slope detection method of FM demodulation.
FM and PM				
9	20-21 2(c)	5	theory	With proper block diagram explain the relation between frequency modulation and phase modulation.
Carson's rule				
10	20-21 4(d)	4	theory	State Carson's rule and its limitations.
11	20-21 TT2 8(e)	5	numerical	Consider an FM signal obtained from a modulating signal frequency of 2000 Hz and maximum Amplitude of 5 volts. The frequency sensitivity of modulator is 2 kHz/V. Find the bandwidth of the FM using Carson's rule.
Pre-emphasis				
12	20-21 TT2 8(d)	5	theory	Explain principle of preemphasis and deemphasis
NBFM				
13	22-23 TT2 7(i)	0	derivation	Derive the narrowband FM ($\beta < 1$) expression.
14	22-23 TT2 7(j)	0	diagram	Draw the NBFM block diagram.
15	21-22 3(a)	5	theory	Explain NBFM generation process with necessary block diagrams and equations.
FM versus AM				
16	22-23 TT2 7(a)	0	theory	Advantages and disadvantages of FM over AM.

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4.5. Receivers:

7 marks, 1 question, in 1 of 4 papers. Book: Lathi Ch. 4.

Most often theory (1 of 1). Subtopics: Superheterodyne.

Formulas:

Subtopic	Formula	Equation	Used in
Superheterodyne (7)	**Local oscillator**	$f_{LO} = f_s + f_{IF}$	20-21 4(a)
	Image frequency	$f_{si} = f_s + 2f_{IF}$	20-21 4(a)

Questions:

No.	Paper	Marks	Type	Question
Superheterodyne				
1	20-21 4(a)	7	theory	With appropriate block diagram, describe the operation of superheterodyne receiver.

4.6. Sampling and pulse modulation:

74 marks, 14 questions, in 3 of 4 papers. Book: Lathi Ch. 6.

Most often theory (7 of 14). Subtopics: Quantization; PCM; Delta modulation; Sampling.

Formulas:

Subtopic	Formula	Equation	Used in
Quantization (4)	Step size	$\Delta = \dfrac{2m_p}{L}$	21-22 4(e), 19-20 4(c)
	Quantization noise	$\dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$	19-20 4(c)
	SQNR (uniform)	$\dfrac{S_o}{N_o} = 3L^2 \dfrac{\overline{m^2}}{m_p^2}$	19-20 4(c)
	μ-law compressor	$y = \dfrac{\ln(1+\mu\lvert m\rvert/m_p)}{\ln(1+\mu)}\operatorname{sgn}(m)$	21-22 4(c), 19-20 4(c)
PCM (32)	Sampling rate	$f_s = (1+x)\,2B$	19-20 4(d)
	PCM bit rate	$R = (1+e)\,N_{sig} f_s n$	19-20 4(d)
	Minimum bandwidth	$B_T = R/2$	19-20 4(d)
	Bits per sample	$n = \log_2 L$	20-21 3(c)
	PCM bit rate	$R_b = n f_s$	21-22 4(e), 20-21 3(c)
	Information rate	$R = r\,H \quad \text{bits/s}$	21-22 4(c)
	Bits per sample	$L = 2^n,\ n = \log_2 L$	21-22 4(e)
	Quantization noise	$N_q = \dfrac{\Delta^2}{12} = \dfrac{m_p^2}{3L^2}$	21-22 4(a)
	SQNR	$\text{SQNR} = 3L^2 \dfrac{\overline{m^2}}{m_p^2}$	21-22 4(a)

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Subtopic	Formula	Equation	Used in
	Peak signal to peak noise	$\frac{S_{peak}}{N_{peak}} = L^2$	21-22 4(e)
	Minimum bandwidth	$B_T = \frac{R_b}{2} = nB$	21-22 4(e)
	Companded SQNR	$\text{SQNR} \approx \frac{3L^2}{[\ln(1+\mu)]^2}$	21-22 4(c)
Delta modulation (26)	DM update	$\hat{m}_q[k] = \hat{m}_q[k-1] + d_q[k],\ d_q = \pm\sigma$	19-20 4(e)
	DM receiver	$\hat{m}_q[k] = \sum_{i=0}^{k} d_q[i]$	19-20 4(e)
	DM largest slope	σf_s	20-21 3(a)
	No slope overload	$\left\lvert \frac{dm}{dt} \right\rvert_{max} \leq \sigma f_s$	20-21 3(a)
Sampling (12)	**Nyquist rate**	$f_s \geq 2B$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_n G(f - nf_s)$	20-21 4(c)
	Interpolation formula	$g(t) = \sum_k g(kT_s)\,\text{sinc}(2\pi Bt - k\pi)$	21-22 4(b), 20-21 4(c)
	Sampled spectrum	$\bar{G}(f) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} G(f - nf_s)$	21-22 4(b)
	Reconstruction LPF	$H(f) = T_s\,\text{rect}\left(\frac{f}{2f_0}\right),\ B \leq f_0 \leq f_s - B$	21-22 4(b)

Symbols:

L = number of levels ↩

μ = compression constant (255), not the AM index ↩

x = fraction above Nyquist ↩

e = framing fraction ↩

n = bits per sample; bits per code word ↩

R_b = bit rate (bit/s) ↩

r = symbol rate (symbols/s) ↩

$\overline{m^2}$ = message power ↩

σ = step size ↩

B = highest message frequency (Hz) ↩

f_s = sampling rate (samples/s) ↩

$G(f)$ = message spectrum

$T_s = 1/f_s$ = sampling interval (s)

f_0 = filter cutoff (Hz) ↩

Questions:

No.	Paper	Marks	Type	Question
Quantization				
1	19-20 4(c)	4	theory	Define and explain uniform and non-uniform quantization.
PCM				
2	21-22 4(a)	5	theory	What is PCM? Explain PCM system.
3	21-22 4(c)	5	theory	Write short note on: Companding, Information rate, and Entropy of a source.
4	21-22 4(e)	5	numerical	An audio signal is sampled at 8 kHz rate using a 32-level quantizer. Find- (i) Peak signal power to peak quantization noise power. (ii) System bandwidth.

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No.	Paper	Marks	Type	Question
5	21-22 TT3 9(a)	7	numerical repeats 20-21 5(c)	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
6	20-21 3(c)	5	diagram	Explain the basic block diagram of PCM system.
7	19-20 4(d)	5	numerical	Five telemetry signals, each of bandwidth 2400 Hz, are to be transmitted simultaneously by binary PCM. The signals must be sampled at least 20% above the Nyquist rate. Framing and synchronizing requires an additional 0.5% extra bits. A PCM encoder is used to convert these signals before they are time-multiplexed into a single data stream. Determine the minimum possible data rate (bit per second) that must be transmitted, and the minimum bandwidth required to transmit the multiplexed signal.
Delta modulation				
8	21-22 5(a)	6	theory repeats 19-20 5(a)	Explain the principle of sigma-delta modulator. Also explain how it is advantageous over DM.
9	20-21 3(a)	5	theory	Define slope overload distortion and granular noise in delta modulation and how it is removed in ADM?
10	20-21 3(b)	5	diagram	Draw the transmitter and receiver circuits block diagram of delta modulation.
11	19-20 4(e)	5	derivation	Show that the receiver of DM is essentially an accumulator.
12	19-20 5(a)	5	theory	Explain principle of sigma-delta modulator. Also explain how it is advantageous over DM.
Sampling				
13	21-22 4(b)	5	theory	”A proper valued LPF filter can be used to reconstruct the sampled signal”-justify the statement with the proper mathematical explanations.
14	20-21 4(c)	7	derivation	State and prove Nyquist sampling theorem. Also find the interpolation formula for the reconstruction of analog signals from its uniform samples.

4.7. Digital modulation:

75 marks, 12 questions, in 4 of 4 papers. Book: Lathi Ch. 7, 10; Forouzan Ch. 5.

Most often theory (8 of 12). Subtopics: Line coding; DPSK; ASK, FSK and PSK; QPSK; FSK; QAM.

Formulas:

Subtopic	Formula	Equation	Used in
Line coding (7)	AMI	$0 \to 0$ V; 1s alternate $\pm V$	19-20 5(b)
	Dicode	$0 \to 1$: $-V$; $1 \to 0$: $+V$; no change: 0	19-20 5(b)
DPSK (22)	**DPSK encoder**	$q_k = I_k \oplus q_{k-1}$	22-23 8(b), 21-22 5(c), 20-21 6(d), 19-20 5(c)
	DPSK receiver	LPF output $= \frac{1}{2} a_k a_{k-1} A^2$	22-23 8(b), 19-20 5(c)
	DPSK receiver	$y_k \propto a_k a_{k-1}$	21-22 5(c), 20-21 6(d)
ASK, FSK and PSK (16)	**BFSK correlator**	$\int_0^{T_b} A \cos^2 \omega_1 t \, dt \approx \frac{AT_b}{2}$	19-20 5(d)

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Subtopic	Formula	Equation	Used in
	Coherent BFSK error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$	19-20 5(d)
	ASK and PSK bandwidth	$B = (1+d)S$	21-22 4(d)
	Coherent BASK output	$dA\cos^2\omega_c t \to \frac{dA}{2}$	21-22 5(d)
QPSK (8)	**QPSK**	$\varphi = a_I A\cos\omega_c t + a_Q A\sin\omega_c t$	19-20 6(a)
	BPSK/QPSK error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	19-20 6(a)
FSK (8)	FSK frequencies	$f_{1,0} = f_c \pm \Delta f$	20-21 6(a)
	FSK bandwidth	$B_{FSK} \approx 2\Delta f + 2R_b$	20-21 6(a)
	FSK bandwidth (Forouzan)	$B = (1+d)S + 2\Delta f$	21-22 4(d), 20-21 6(a)
QAM (14)	**QAM signal**	$\varphi_{QAM} = m_1\cos\omega_c t + m_2\sin\omega_c t$	20-21 6(b)
	QAM with phase error	$m_1\cos\theta - m_2\sin\theta$	20-21 6(b)
	QAM signal	$\varphi_{QAM}(t) = m_1(t)\cos\omega_c t + m_2(t)\sin\omega_c t$	21-22 5(b)
	QAM with phase error	$m_1(t)\cos\theta - m_2(t)\sin\theta$	21-22 5(b)

Symbols:

I_k = data bit ↩

q_k = encoded bit ↩

a_k = polar symbol (±1) ↩

y_k = LPF output ↩

S = signal (baud) rate; baud rate ↩

d = filter factor, $0 \le d \le 1$; data (1 or 0) ↩

R_b = bit rate ↩

Δf = half the spacing of f_1 and f_2 (Hz) ↩

m_1, m_2 = the two messages ↩

Questions:

No.	Paper	Marks	Type	Question
Line coding				
1	19-20 5(b)	7	diagram	Briefly explain the desired characteristics of a line coding scheme. Draw the line coded signal for the information bits '1010011100' for the following line coding scheme- i) Manchester, ii) AMI, iii) Di-code.
DPSK				
2	22-23 TT3 8(b)	4	diagram repeats 20-21 6(d)	Draw the block diagram of a Differential PSK (DPSK) modulator and demodulator and show how the I_k sequence [1010011100] changes in different parts of the block.
3	21-22 5(c)	6	numerical repeats 20-21 6(d)	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010010110'.
4	20-21 6(d)	4	numerical	Tabulate the values of signal at each stage of DPSK encoder and receiver for message signal '1010011100'
5	19-20 5(c)	8	theory	Explain DPSK principle with necessary block diagram.
ASK, FSK and PSK				
6	21-22 4(d)	5	theory	Name three mechanisms for the modulation of digital data into an analog signal. Show the differences among them.
7	21-22 5(d)	6	theory	Briefly describe coherent and non-coherent detection of BASK and BFSK.
8	19-20 5(d)	5	theory	Describe coherent detection of BFSK.

No.	Paper	Marks	Type	Question
QPSK				
9	19-20 6(a)	8	theory	With proper block diagram, describe modulation of QPSK. Also explain how it offers advantages over BPSK?
FSK				
10	20-21 6(a)	8	theory	Briefly describe FSK generation with the necessary block diagram. Include its spectrum and discuss the bandwidth requirements, along with the advantages and disadvantages of FSK.
QAM				
11	21-22 5(b)	7	theory repeats 20-21 6(b)	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.
12	20-21 6(b)	7	theory	With proper block diagram explain the principle of coherent detection of QAM. Mention its advantages and drawbacks.

4.8. Multiplexing and multiple access:

82 marks, 15 questions, in 4 of 4 papers. Book: Lathi Ch. 11; Forouzan Ch. 6, 12.

Most often theory (9 of 15). Subtopics: TDM; FDM; Multiple access; Spread spectrum.

Formulas:

Subtopic	Formula	Equation	Used in
TDM (50)	**Frame rate**	$\dfrac{\text{channel rate}}{\text{bits per slot}}$	19-20 6(b)
	Frame and bit duration	$T_f = 1/\text{frame rate},\ T_b = 1/R$	19-20 6(b)
	Link rate	$R = \text{frame rate} \times \text{bits per frame}$	19-20 6(b)
	Pulse stuffing	pad slow inputs to the fastest rate	19-20 6(d)
	Multiple slots	slots per frame $= R_i/R_{base}$	19-20 6(d)
	Quantization step	$\Delta = \dfrac{2m_p}{L}$	20-21 5(c)
	Largest error	$e_{max} = \dfrac{\Delta}{2} = \dfrac{m_p}{L}$	20-21 5(c)
	TDM bit rate	$R_b = N f_s n$	21-22 6(b), 20-21 5(c)
	Minimum bandwidth	$B_T = \dfrac{R_b}{2}$	20-21 5(c)
	Levels from an error limit	$\dfrac{m_p}{L} \le \epsilon\, m_p \ \Rightarrow\ L \ge \dfrac{1}{\epsilon}$	21-22 6(b)
	Synchronous TDM link	$R_{link} = \sum R_i$	21-22 6(a)
	Character TDM	frame $= 4 \times 8 + 1 = 33$ bits; link $= 250 \times 33 = 8250$ bit/s	21-22 9(c)
	Pulse stuffing	slow channel padded to the fast rate	22-23 8(c)
	TDM frame	frame rate = channel rate / bits per slot; R = frame rate × bits per frame	22-23 8(c)
FDM (10)	**FDM bandwidth**	$B_{total} = NB_{ch} + (N-1)B_g$	19-20 6(c)
	FDM link bandwidth	$B_{link} = NB_{ch} + (N-1)B_g$	20-21 2(b)

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Subtopic	Formula	Equation	Used in
Spread spectrum (6)	FHSS channels	$M = 2^k$, $B_{SS} \approx MB$	20-21 6(c)

Symbols:

m_p = peak signal ↩

N = number of channels ↩

ϵ = allowed error as a fraction of m_p ↩

R_i = input rates ↩

B_g = guard band ↩

k = PN pattern bits

Questions:

No.	Paper	Marks	Type	Question
TDM				
1	22-23 TT3 8(c)	6	numerical	Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse stuffing TDM with no synchronization bits. Answer the following questions: (a) What is the size of a frame in bits? (b) What is the frame rate? (c) What is the duration of a frame? (d) What is the data rate?
2	21-22 6(a)	6	theory	Write short notes on: Multilevel multiplexing and Multiple slot allocation.
3	21-22 6(b)	6	numerical repeats 20-21 5(c)	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patients are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitudes is 0.25 % of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
4	21-22 TT3 9(b)	6	theory repeats 21-22 6(a)	Write short notes in multilevel multiplexing and multiple slot allocation.
5	21-22 TT3 9(c)	7	numerical	We have four sources, each creating 250 characters (8 bits per character) per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find (a) the data rate of each source, (b) the duration of each character in each source, (c) the frame rate, (d) the duration of each frame, (e) the number of bits in each frame, and (f) the data rate of the link.
6	20-21 5(c)	7	numerical	It is desired to set up a central station for simultaneous monitoring of the electrocardiograms (ECGs) of 10 hospital patients. The data from the 10 patents *[sic: should be "patients"]* are brought to a processing center over wires and are sampled, quantized, binary-coded, and time-division-multiplexed. The multiplexed data are now transmitted to the monitoring station. The ECG signal bandwidth is 100 Hz. The maximum acceptable error in sample amplitude is 0.25% of the peak signal amplitude. The sampling rate must be at least twice the Nyquist rate. Determine the minimum cable bandwidth needed to transmit these multiplexed data.
7	19-20 6(b)	6	numerical	A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output for the following inputs- '110010', '001010', '101101', '000111'. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?
8	19-20 6(d)	6	theory	Write short notes on following terms- (i) Pulse stuffing (ii) Multi-slot multiplexing
FDM				
9	20-21 2(b)	5	numerical	Five channels, each with 100 kHz bandwidth, are to be frequency division multiplexing together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channel to prevent interference?
10	19-20 6(c)	5	theory	Briefly explain frequency division multiplexing techniques.

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No.	Paper	Marks	Type	Question
Multiple access				
11	21-22 6(c)	3	theory	Compare between FDMA and CDMA.
12	21-22 6(e)	3	theory repeats 20-21 2(a)	Differentiate between multiplexing and multiple access technologies.
13	20-21 2(a)	5	theory	Differentiate between multiplexing and multiple access technologies.
14	20-21 5(a)	5	theory	Write down the comparison between TDMA and CDMA.
Spread spectrum				
15	20-21 6(c)	6	theory	Explain the principle of frequency hopping spread spectrum technique.

4.9. Information theory:

29 marks, 5 questions, in 3 of 4 papers. Book: Lathi Ch. 13.

Most often numerical (3 of 5). Subtopics: Entropy.

Formulas:

Subtopic	Formula	Equation	Used in
Entropy (29)	Information in a message	$I_i = \log_2 \frac{1}{P_i}$ bits	19-20 2(a)
	Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/message	19-20 2(a)
	Count in a long sequence	$N_i = NP_i$	19-20 2(a)
	Information rate	$R = rH$ bits/s	19-20 2(a)
	Information in a symbol	$I = \log_2 \frac{1}{P}$ bits	20-21 1(a)
	Entropy	$H = \sum_i P_i \log_2 \frac{1}{P_i}$ bits/symbol	20-21 1(a)
	Binary entropy	$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$	21-22 1(a), 20-21 1(a)
	Information in a symbol	$I(m_i) = \log_2 \frac{1}{P_i}$ bits	21-22 1(a)
	Entropy	$H(m) = \sum_{i=1}^{n} P_i \log_2 \frac{1}{P_i}$ bits/symbol	21-22 1(a)
	Bounds of entropy	$0 \le H(m) \le \log_2 n$	21-22 1(a)
	Rate of a four-message source	$H = 1.846$ bits, $R = rH = 1.846$ Mbit/s	21-22 7(a)

Symbols:

P_i = probability of message m_i; probability of symbol m_i ↩

N = number of messages sent

r = messages per second

P = probability of the symbol ↩

H = average information per symbol ↩

p = probability of one symbol; probability of one of two symbols ↩

n = number of symbols ↩

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Questions:

No.	Paper	Marks	Type	Question
Entropy				
1	21-22 1(a)	6	numerical	Find the entropy $H(m)$ of a memoryless source.
2	21-22 TT1 7(a)	5	numerical	A message source generates one of the four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, and 0.1. Each emitted message is independent of the other messages in the sequence: (i) Find the source entropy. (ii) Find the rate of information generated by this source (in bits per second).
3	20-21 1(a)	5	numerical	Find the entropy of single bit binary memoryless source. Also draw the curve of entropy Vs. probability for that source.
4	20-21 TT1 7(a)	5	diagram repeats 20-21 1(a)	Find the entropy of single bit binary memoryless source. Also draw the curve of entropy vs probability for that source.
5	19-20 2(a)	8	derivation	Show that the average information per message is equal to the entropy of the information source.

4.10. System design:

7 marks, 1 question, in 1 of 4 papers. Book: Lathi Ch. 1.

Most often theory (1 of 1). Subtopics: Link design.

Questions:

No.	Paper	Marks	Type	Question
Link design				
1	21-22 6(d)	7	theory	Discuss the design procedures of a communication system.

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Signals and spectrum				
1	21-22 1(b)	7	Propagation modes	Ground wave below 2 MHz follows the Earth (AM); sky wave 2 to 30 MHz is bent back by the ionosphere (long range); line of sight above 30 MHz, range $\approx 4.12(\sqrt{h_t}+\sqrt{h_r})$ km (FM, TV, mobile, satellite).
2	20-21 1(b)	6	EM spectrum	Radio: 3 kHz to 1 GHz, ground, sky or line of sight, omnidirectional, broadcasting. Microwaves: 1 to 300 GHz, line of sight, directional, satellite and cellular. Infrared: 300 GHz to 400 THz, short range, blocked by walls, remote controls.
3	19-20 1(a)	6	Transmission media	Twisted pair: two insulated wires twisted so interference cancels in their difference; UTP and STP, categories; telephone and LAN; bandwidth and distance limited. Radio waves: 3 kHz to 1 GHz, omnidirectional, ground, sky or line of sight, pass through walls; AM, FM and TV.
Noise and channel capacity				
4	20-21 1(c)	4	Thermal and transit-time noise	Thermal: random electron motion from heat, $P_n = kTB$, white, in every resistor. Transit time: at very high frequency the carrier crossing time is comparable with the signal period, so the noise rises with frequency.
Amplitude modulation				
5	22-23 TT2 7(c)	0	DSB-SC	A switching modulator multiplies $m(t)$ by a ± 1 square wave at f_c; in the ring modulator a large carrier switches the two diode pairs; output $m(t)w(t)$, and a BPF keeps $(4/\pi)m\cos\omega_c t$; with $m = 0$ the balanced currents cancel, so no carrier.

No.	Paper	Marks	Subtopic	Points to cover
6	22-23 TT2 7(d)	0	SSB	AM: carrier and two sidebands, $2B$, envelope detector, at most 33 % efficient. DSB-SC: $2B$, coherent detector, 100 %. SSB-SC: B, coherent detector, 100 %, hardest to generate.
7	22-23 TT2 7(f)	0	SSB	Weaver: two quadrature stages with LPFs between them; no sharp filter and no wideband shifter, but two pairs of balanced modulators, exact quadrature at both stages and matched LPFs; any imbalance leaves part of the unwanted sideband and a tone at the band centre; complex.
8	22-23 TT2 7(g)	0	VSB	VSB: one full sideband plus a vestige of the other; the filter has complementary symmetry about f_c, so the detected output is flat; used when the message has power near dc (video) and SSB filtering is impossible, with less bandwidth than DSB.
9	22-23 TT2 7(h)	0	VSB	Video has power down to dc, so SSB is not possible; DSB wastes bandwidth; VSB saves about 40 %; with a carrier (WC) a cheap envelope detector works in every TV set.
10	21-22 2(b)	6	DSB-SC	A local carrier with phase error θ and frequency error $\Delta\omega$ gives $m\cos(\Delta\omega t+\theta)$: a phase error attenuates (null at 90°), a frequency error beats; so the carrier must be synchronized (pilot, Costas loop).
11	21-22 2(c)	8	DSB-SC	The I branch multiplies by cos, the Q branch by sin from a VCO; the LPF outputs are $\frac{1}{2}m\cos\theta$ and $\frac{1}{2}m\sin\theta$; their product $\frac{1}{8}m^2\sin 2\theta$, through a narrow LPF, steers the VCO to $\theta=0$; the I output is the message.
12	21-22 3(b)	5	Envelope detector	Diode and a parallel RC: C charges to each carrier peak and discharges slowly through R, so the output follows $A+m(t)$; needs $1/\omega_c \ll RC < 1/(2\pi B)$ and $\mu \le 1$; too large an RC clips diagonally, too small leaves ripple; a series C removes the dc (levelling).
13	21-22 3(d)	10	AM modulation index	$A=m_p/\mu$; draw the envelope $A+m(t)$ and its mirror; $\mu>1$ overmodulates and the envelope detector fails.
14	21-22 TT2 8(a)	7	DSB-SC	Quadrature null: the coherent output $\frac{1}{2}m\cos\theta$ vanishes at $\theta=90°$. Costas: I and Q branches; their product $\frac{1}{8}m^2\sin 2\theta$ steers the VCO to $\theta=0$, so the carrier stays locked.
15	20-21 5(d)	0	DSB-SC	The product modulator shifts $M(f)$ to $\pm f_c$; the coherent detector shifts it back and the LPF keeps $(A_c/2)M(f)$; if $f_c<W$ the shifted copies overlap near baseband and the output is distorted; so $f_c \ge W$.
16	20-21 TT2 8(a)	5	DSB-SC	DSB-WC: carrier impulses at $\pm\omega_c$ plus both sidebands (the message spectrum shifted), bandwidth $2\omega_m$. SSB-SC: only the upper (or lower) half, no carrier, bandwidth ω_m.
17	19-20 2(d)	3	SSB	A real filter needs a transition band; the gap between the USB and the LSB is $2f_L$; so the message must have no power near dc (speech has none below 300 Hz: a 600 Hz gap); otherwise use the phase-shift method or VSB.
Angle modulation				
18	22-23 TT2 7(a)	0	FM versus AM	FM advantages: better noise immunity, constant envelope (efficient amplifiers), capture effect, higher fidelity. Disadvantages: much wider bandwidth, line-of-sight range, complex equipment, threshold effect.
19	22-23 TT2 7(j)	0	NBFM	Integrator, balanced modulator fed with the carrier shifted by $-90°$, and a summer adding the carrier.
20	21-22 3(a)	5	NBFM	$\varphi_{NBFM} \approx A\cos\omega_c t - Ak_f a(t)\sin\omega_c t$ with $a=\int m\,dt$: integrator, balanced modulator fed with the carrier shifted by $-90°$, subtract from the carrier; valid for $\beta \ll 1$.
21	20-21 2(c)	5	FM and PM	$\omega_i = d\theta/dt$; FM of m is PM of $\int m\,dt$, and PM of m is FM of $\dot{m}$; an integrator before a phase modulator gives FM, a differentiator before a frequency modulator gives PM.
22	20-21 4(d)	4	Carson's rule	$B_T \approx 2(\Delta f + f_m) = 2f_m(\beta+1)$, holding about 98 % of the power. Limits: only approximate; underestimates for $2<\beta<10$; a general message needs its highest f_m; weak lines are ignored.

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No.	Paper	Marks	Subtopic	Points to cover
23	20-21 TT2 8(d)	5	Pre-emphasis	FM output noise rises as f^2 while the high audio frequencies are weak; boost them before modulation (pre-emphasis, 75 μs) and cut them equally after demodulation (de-emphasis), $H_pH_d = 1$; about 13 dB better SNR.
24	19-20 3(b)	8	FM bandwidth	Amplitude sets $\Delta f = k_f A_m$: more significant lines, bandwidth rises nearly in proportion. Frequency sets the line spacing f_m and lowers β: fewer lines, bandwidth rises only a little (Carson).
25	19-20 4(b)	6	FM demodulation	Limiter removes amplitude changes; a tuned circuit tuned above f_c puts the carrier on its slope, so frequency changes become amplitude changes (a differentiator); an envelope detector follows; one slope is not linear, so the balanced version uses two tuned circuits.
Receivers				
26	20-21 4(a)	7	Superheterodyne	RF amplifier, mixer with $f_{LO} = f_s + f_{IF}$, fixed IF amplifier (455 kHz), detector, audio amplifier, AGC; the image is at $f_s + 2f_{IF}$; the fixed IF gives good selectivity and sensitivity.
Sampling and pulse modulation				
27	21-22 4(a)	5	PCM	Sampling ($f_s \geq 2B$), quantizing (L levels), encoding ($n = \log_2 L$ bits); a line code over the channel with regenerative repeaters; the receiver regenerates, decodes and low-pass filters; $R_b = nf_s$.
28	21-22 4(b)	5	Sampling	Sampled spectrum $\bar{G}(f) = (1/T_s)\sum G(f - nf_s)$; with $f_s \geq 2B$ the copies do not overlap; an LPF of gain T_s and cutoff between B and $f_s - B$ keeps $G(f)$; in time this is the sinc interpolation formula.
29	21-22 4(c)	5	PCM	Companding: compress, quantize uniformly, expand ($\mu = 255$, A = 87.6) for a nearly constant SQNR. Information rate $R = rH$. Entropy $H = \sum P_i \log_2(1/P_i)$, largest for equal probabilities.
30	21-22 5(a)	6	Delta modulation	Integrate the message, then delta-modulate (one integrator after the subtractor); the pulse density follows the amplitude; the receiver is only an LPF. Over DM: no frequency-dependent slope overload, no error build-up, dc passes, simpler receiver, noise shaping with oversampling.
31	20-21 3(a)	5	Delta modulation	Slope overload: the signal slope exceeds σf_s and the staircase lags. Granular noise: the staircase hunts by $\pm\sigma$ on a flat signal. ADM raises the step on runs of equal bits and lowers it on alternating bits.
32	20-21 3(b)	5	Delta modulation	Transmitter: subtractor, one-bit quantizer, accumulator in the feedback path. Receiver: accumulator, then LPF.
33	20-21 3(c)	5	PCM	Anti-aliasing LPF, sampler, quantizer, encoder; channel with regenerative repeaters; regenerator, decoder, reconstruction LPF; $R_b = nf_s$.
34	19-20 4(c)	4	Quantization	Uniform: equal steps $\Delta = 2m_p/L$, constant noise $\Delta^2/12$, so weak signals get a poor SQNR. Non-uniform: small steps near zero, large at the peaks, nearly constant SQNR; done by companding (μ-law, A-law).
35	19-20 5(a)	5	Delta modulation	Integrate the message, then delta-modulate (one integrator after the subtractor); the pulse density follows the amplitude; the receiver is only an LPF. Over DM: no frequency-dependent slope overload, no error build-up, dc passes, simpler receiver, noise shaping with oversampling.
Digital modulation				
36	22-23 TT3 8(b)	4	DPSK	$q_k = I_k \oplus q_{k-1}$ through a one-bit delay, polar coder, carrier multiplier; the receiver multiplies by the one-bit-delayed signal, low-pass filters and reads the sign.
37	21-22 4(d)	5	ASK, FSK and PSK	ASK (amplitude), FSK (frequency), PSK (phase); bandwidth: ASK and PSK about $2R_b$, FSK more; in noise: PSK best, ASK worst; ASK simplest to build.

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No.	Paper	Marks	Subtopic	Points to cover
38	21-22 5(b)	7	QAM	$\varphi = m_1 \cos\omega_c t + m_2 \sin\omega_c t$; multiply by $2\cos\omega_c t$ and $2\sin\omega_c t$, low-pass filter to get m_1 and m_2. Advantage: two signals in one band. Drawbacks: needs an exact carrier; a phase error gives crosstalk $m_1\cos\theta - m_2\sin\theta$; needs linear amplifiers.
39	21-22 5(d)	6	ASK, FSK and PSK	BASK: coherent = multiply by the carrier, integrate, threshold; non-coherent = BPF, envelope detector, threshold. BFSK: coherent = two correlators and compare; non-coherent = two BPFs with envelope detectors and compare. Coherent is better in noise but needs carrier sync.
40	20-21 6(a)	8	FSK	Bits select $f_c \pm \Delta f$: two switched oscillators or a VCO; the spectrum is two ASK lobes, $B \approx 2\Delta f + 2R_b$. Advantages: constant envelope, simple, noise immune. Disadvantages: wide bandwidth, worse than PSK.
41	20-21 6(b)	7	QAM	$\varphi = m_1 \cos\omega_c t + m_2 \sin\omega_c t$; multiply by $2\cos\omega_c t$ and $2\sin\omega_c t$, low-pass filter to get m_1 and m_2. Advantage: two signals in one band. Drawbacks: needs an exact carrier; a phase error gives crosstalk $m_1\cos\theta - m_2\sin\theta$; needs linear amplifiers.
42	19-20 5(b)	7	Line coding	Desired: small bandwidth, no dc, timing content, power efficiency, error detection, transparency.
43	19-20 5(c)	8	DPSK	$q_k = I_k \oplus q_{k-1}$, then PSK: a 1 flips the phase; the receiver multiplies by the signal delayed by T_b, low-pass filters and reads the sign; no local carrier; about 1 dB worse than PSK; errors come in pairs.
44	19-20 5(d)	5	ASK, FSK and PSK	Two correlators with $\cos\omega_1 t$ and $\cos\omega_2 t$, integrated over T_b; the matching branch gives $AT_b/2$, the other about 0; a comparator decides; both carriers must be in sync.
45	19-20 6(a)	8	QPSK	Serial to parallel; odd bits on $\cos\omega_c t$, even bits on $\sin\omega_c t$, each lasting $2T_b$; the sum has four phases. Over BPSK: half the bandwidth for the same bit rate, the same bit error rate.
Multiplexing and multiple access				
46	21-22 6(a)	6	TDM	Multilevel: merge slow inputs first into one faster stream, then multiplex it with the fast ones. Multiple slot: a fast input gets several slots per frame in proportion to its rate. Both need rates that are integer multiples.
47	21-22 6(c)	3	Multiple access	FDMA: each user its own band, guard bands, simple, hard capacity limit. CDMA: same band and time, separated by codes, soft capacity, needs power control, resists jamming and multipath.
48	21-22 6(e)	3	Multiple access	Multiplexing: several signals combined at one point onto one link (FDM, TDM, WDM), under one owner. Multiple access: separate users share one medium (FDMA, TDMA, CDMA) and need access rules.
49	21-22 TT3 9(b)	6	TDM	Multilevel: merge slow inputs first into one faster stream, then multiplex it with the fast ones. Multiple slot: a fast input gets several slots per frame in proportion to its rate. Both need rates that are integer multiples.
50	20-21 2(a)	5	Multiple access	Multiplexing: several signals combined at one point onto one link (FDM, TDM, WDM), under one owner. Multiple access: separate users share one medium (FDMA, TDMA, CDMA) and need access rules.
51	20-21 5(a)	5	Multiple access	TDMA: users take turns in time slots on one frequency, need exact timing, hard capacity limit. CDMA: users share the band at the same time, separated by codes, need power control, soft capacity, resist jamming.
52	20-21 6(c)	6	Spread spectrum	A PN code drives a frequency synthesizer, so the carrier hops among 2^k frequencies; the receiver hops in step; $B_{SS} \approx MB$; slow and fast hopping; resists jamming and eavesdropping.

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No.	Paper	Marks	Subtopic	Points to cover
53	19-20 6(c)	5	FDM	Each signal modulates its own carrier; the bands sit side by side with guard bands on one link; the receiver separates them with BPFs and demodulates; hierarchy: group, supergroup, master group; radio, TV and telephone.
54	19-20 6(d)	6	TDM	Pulse stuffing: dummy bits pad slower inputs up to the fastest when the rates are not integer multiples; the receiver drops them. Multiple slot: an input at a multiple of the base rate gets that many slots per frame.
Information theory				
55	20-21 TT1 7(a)	5	Entropy	Write $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$; zero at $p = 0$ and 1, maximum 1 bit at $p = 0.5$; draw the symmetric arch.
System design				
56	21-22 6(d)	7	Link design	Requirements; channel study; Shannon limit; source coding; channel coding; modulation; multiplexing and multiple access; link budget; trade bandwidth, power and error rate.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Signals and spectrum				
1	21-22 1(d)	6	Distortionless transmission	Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions.
2	20-21 TT1 7(b)	5	Distortionless transmission	Write the output as a delayed scaled input; Fourier transform it; read magnitude and phase conditions.
Noise and channel capacity				
3	21-22 TT1 7(b)	5	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.
4	20-21 TT1 7(d)	5	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.
5	19-20 1(d)	7	Cascaded noise	Model each stage as a noiseless gain with a noise resistor; add output noise powers; divide by the total squared gain.
Amplitude modulation				
6	22-23 TT2 7(e)	0	DSB-SC	Multiply DSB-SC and SSB by a carrier with frequency and phase error; low-pass filter; read attenuation, beating, phase and frequency shift.
7	21-22 2(d)	6	VSB	Write the VSB spectrum; shift by the carrier; keep the baseband terms; require a constant bracket.
8	21-22 TT1 7(d)	5	Envelope detector	Compare the capacitor discharge slope with the envelope slope; find the worst instant; get the RC bound.
9	21-22 TT2 8(b)	6	SSB	Add a large carrier to SSB; write the envelope; expand the square root for a large carrier.
10	21-22 TT2 8(c)	7	VSB	Write the four DSB lines; apply the filter gains 1, a, 1-a, 0; demodulate; the partial sidebands add to the full tone.
11	20-21 4(b)	7	VSB	Shape DSB-SC with a filter; multiply by the carrier; filter; require the sum of the shifted responses to be constant.
12	20-21 TT2 8(b)	5	DSB-SC	Multiply by a carrier with frequency and phase error; low-pass filter; read the cos factor.
13	19-20 2(c)	6	DSB-SC	Multiply by a carrier with frequency and phase error; low-pass filter; read the cos factor.
14	19-20 3(a)	8	SSB	Two branches $m\cos\omega_c t$ and $m_h \sin\omega_c t$; tone case with product identities; subtract or add to cancel a sideband.

No.	Paper	Marks	Derivation	Outline
Angle modulation				
15	22-23 TT2 7(i)	0	NBFM	Expand cos of the FM angle; use small-angle approximations; get carrier minus quadrature sideband term.
Sampling and pulse modulation				
16	20-21 4(c)	7	Sampling	Multiply by an impulse train; write its Fourier series; read off the repeated spectrum; filter with an ideal LPF.
17	19-20 4(e)	5	Delta modulation	Write the DM update; unroll it; the estimate is the running sum of the steps.
Information theory				
18	19-20 2(a)	8	Entropy	Count each message NP_i times in N messages; add their information; divide by N.

7. Design recipes:

Every design asked, in book order. Each recipe gives the steps in the order to take them; the link opens the full working.

No.	Paper	Marks	Design	Recipe
Angle modulation				
1	22-23 TT3 8(a)	10	Armstrong modulator	Total multiplication $n_1 n_2 = \Delta f / \Delta f_1$ (all doublers, so a power of 2); for each split find $f_{LO} = \lvert n_1 f_{c1} \pm f_c / n_2 \rvert$ and keep the one inside the LO range; draw the chain with f_c and Δf after every block. Here $4096 = 128 \times 32$, LO 480.625 kHz (sum).
2	19-20 3(c)	9	Armstrong modulator	Total multiplication $n_1 n_2 = \Delta f / \Delta f_1$ (all doublers, so a power of 2); for each split find $f_{LO} = \lvert n_1 f_{c1} \pm f_c / n_2 \rvert$ and keep the one inside the LO range; draw the chain with f_c and Δf after every block. Here $2048 = 16 \times 128$, LO 440.15625 kHz (sum).

8. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Noise and channel capacity					
1	21-22 1(c)	6	Channel capacity	Attenuation $20\log(V_s/V_r)$; SNR $= (V_r/V_n)^2$; Shannon $C = B\log_2(1+\text{SNR})$; for 100 Mbps solve SNR $= 2^{R/B} - 1$, then the voltage.	13.98 dB; 86.44 Mbps; 5.12 V received (25.6 V sent)
2	20-21 3(d)	10	Channel capacity	$f_s = 1.25 \times 2B$; $n = \log_2 L$; $R = nf_s$; Shannon $C = B\log_2(1+S/N)$ and compare; solve for the S/N or the B that gives $C = R$.	80 kbit/s; no ($C = 66.6$ kbit/s); $S/N = 255$ (24.1 dB); 12.02 kHz
3	20-21 5(b)	6	Channel capacity	dB to ratio; $C = B\log_2(1+\text{SNR})$; for 4.8 kbps, SNR $= 2^{R/B} - 1$.	33.89 kbit/s; SNR 1.66 (2.2 dB)
4	20-21 TT1 7(c)	0	Channel capacity	Attenuation $20\log(V_s/V_r)$; SNR $= (V_r/V_n)^2$; Shannon $C = B\log_2(1+\text{SNR})$; for 100 Mbps solve SNR $= 2^{R/B} - 1$, then the voltage.	13.98 dB; 86.44 Mbps; 5.12 V received (25.6 V sent)
5	19-20 1(c)	6	Channel capacity	$C = B\log_2(1+\text{SNR})$; for double the rate, $1+\text{SNR}' = (1+\text{SNR})^2$.	$C \approx 10$ kbps; SNR ≈ 100.6 (20 dB)

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No.	Paper	Marks	Subtopic	Method	Answer
Amplitude modulation					
6	22-23 TT2 7(b)	0	Sidebands and power	$\mu = V_m/V_c$; $f_c \pm f_m$; side amplitude $\mu V_c/2$; $B = 2f_m$; $P_t = (V_c^2/2R)(1 + \mu^2/2)$; $\eta = \mu^2/(2 + \mu^2)$.	$\mu = 0.2$; 100.5 and 99.5 kHz; 5 V each; 1 kHz; 2.125 W; 1.96 %
7	21-22 2(a)	5	AM modulation index	$I_t = I_c\sqrt{1 + \mu^2/2}$; solve for μ; then put $\mu = 0.8$.	$\mu \approx 70$ %; 9.19 A at $\mu = 0.8$
8	21-22 TT1 7(c)	5	AM modulation index	$I_t = I_c\sqrt{1 + \mu^2/2}$; solve for μ; then put $\mu = 0.8$.	$\mu \approx 70$ %; 9.19 A at $\mu = 0.8$
9	20-21 1(d)	4	AM modulation index	$\mu = V_m/V_c$; side frequencies $f_c \pm f_m$; each side amplitude $\mu V_c/2$.	$\mu = 0.6$; 10.005 and 9.995 MHz; 3 V each
10	20-21 1(e)	6	Sidebands and power	$\mu = V_m/V_c$; side bands $f_c \pm f_m$; $P_t = (V_c^2/2R)(1 + \mu^2/2)$.	$\mu = 0.2$; 100.5 and 99.5 kHz; 2.125 W
11	20-21 TT2 8(c)	5	AM modulation index	$I_t = I_c\sqrt{1 + \mu^2/2}$; solve for μ; then put $\mu = 0.8$.	$\mu \approx 70$ %; 9.19 A at $\mu = 0.8$
12	19-20 1(b)	6	DSB-SC	Expand each message into tones ($\sin^2 = (1 - \cos)/2$, product to sum); a tone $a\cos\omega_m t$ gives lines of $a/2$ at $f_c \pm f_m$; a dc part gives a line at f_c.	(i) 1 at 100 kHz, 0.5 at 99.94 and 100.06 kHz; (ii) 1 at 100 kHz, -0.5 at 99.9 and 100.1 kHz; (iii) 0.75 at 99.363, 99.682, 100.318, 100.637 kHz
Angle modulation					
13	21-22 3(c)	5	Deviation and index	Carrier swing $2\Delta f$; highest and lowest $f_c \pm \Delta f$; $\beta = \Delta f/f_m$.	80 kHz; 93.16 to 93.24 MHz; $\beta = 8$
14	20-21 2(d)	10	Deviation and index	$k_f = \Delta f/V_m$; new $\Delta f = k_f V_m$; $\beta = \Delta f/f_m$.	$\Delta f = 5$, 15, 20 kHz; $\beta = 10$, 30, 80
15	20-21 TT2 8(e)	5	Carson's rule	$\Delta f = k_f A_m$; $\beta = \Delta f/f_m$; $B_T = 2(\Delta f + f_m)$.	$\Delta f = 10$ kHz, $\beta = 5$, 24 kHz
16	19-20 2(b)	8	Deviation and index	Power $A^2/2$; instantaneous frequency from the derivative of the angle, adding the peak terms; $\beta = \Delta f/f_m$ with the highest f_m; Carson.	$P = 72$ W, $\Delta f = 11.5$ kHz, $\beta \approx 7.67$, $B_T \approx 26$ kHz
17	19-20 4(a)	5	Deviation and index	$k_f = \Delta f/V_m$; new $\Delta f = k_f V_m$; $\beta = \Delta f/f_m$.	$\beta = 12.5$; (i) 15 kHz, 37.5; (ii) 20 kHz, 100
Sampling and pulse modulation					
18	21-22 4(e)	5	PCM	$S/N_{peak} = L^2$; $n = \log_2 L$; $R = nf_s$; $B_T = R/2$.	1024 (30.1 dB); 40 kbit/s and 20 kHz
19	21-22 TT3 9(a)	7	PCM	$f_s = 2B$ (with the margin given); levels L from the error limit, $n = \lceil \log_2 L \rceil$; $R = 10nf_s$; $B_T = R/2$.	$f_s = 400$ Hz; $n = 9$; 36 kbit/s; 18 kHz
20	19-20 4(d)	5	PCM	$f_s = 1.2 \times 2B$ per signal; times 5 signals and n bits per sample; add 0.5 %; $B_T = R/2$.	$R \approx 231.6$ kbps, $B_T \approx 115.8$ kHz (8 bits assumed)
Digital modulation					
21	21-22 5(c)	6	DPSK	$q_k = I_k \oplus q_{k-1}$ from a reference bit; map 1 to phase 0 and 0 to π; at the receiver the sign of $a_k a_{k-1}$ gives the bit (+ is 0, − is 1).	$q_k = 1100011011$ (reference 0); output 1010010110
22	20-21 6(d)	4	DPSK	$q_k = I_k \oplus q_{k-1}$ from a reference bit; map 1 to phase 0 and 0 to π; at the receiver the sign of $a_k a_{k-1}$ gives the bit (+ is 0, − is 1).	$q_k = 1100010111$ (reference 0); output 1010011100

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No.	Paper	Marks	Subtopic	Method	Answer
Multiplexing and multiple access					
23	22-23 TT3 8(c)	6	TDM	Stuff the slower channel up to the faster rate; one bit per channel per frame; frame rate = 190 000/s; frame duration its inverse; link rate = frame bits × frame rate.	2 bits; 190 000 frames/s; 5.26 μs; 380 kbps (10 kbps stuffed)
24	21-22 6(b)	6	TDM	$f_s = 2B$ (with the margin given); levels L from the error limit, $n = \lceil \log_2 L \rceil$; $R = 10nf_s$; $B_T = R/2$.	$f_s = 400$ Hz; $n = 9$; 36 kbit/s; 18 kHz
25	21-22 TT3 9(c)	7	TDM	Source rate = 250 × 8; character duration 1/250 s; one character per source per frame, so 250 frames/s; frame bits $4 \times 8 + 1$; link rate = frame bits × frame rate.	2 kbps; 4 ms; 250 frames/s; 4 ms; 33 bits; 8250 bps
26	20-21 2(b)	5	FDM	Add the channel bandwidths and the $N - 1$ guard bands between them.	540 kHz
27	20-21 5(c)	7	TDM	$f_s = 2B$ (with the margin given); levels L from the error limit, $n = \lceil \log_2 L \rceil$; $R = 10nf_s$; $B_T = R/2$.	$f_s = 400$ Hz; $n = 9$; 36 kbit/s; 18 kHz
28	19-20 6(b)	6	TDM	Take 2 bits from each channel in turn; frame rate = channel rate / 2; frame duration its inverse; link rate = 4 × 100 kbps; bit duration its inverse.	frames 11001000, 00101101, 10100111; 50 000 frames/s; 20 μs; 400 kbps; 2.5 μs
Information theory					
29	21-22 1(a)	6	Entropy	Information of a message $\log_2(1/P_i)$; average it over the probabilities.	$H = \sum P_i \log_2(1/P_i)$ bits/symbol; $0 \le H \le \log_2 n$
30	21-22 TT1 7(a)	5	Entropy	$H = \sum P_i \log_2(1/P_i)$; $R = rH$ with $r = 10^6$ messages/s.	1.846 bits/message; 1.846 Mbit/s
31	20-21 1(a)	5	Entropy	$H(p) = -p \log_2 p - (1-p) \log_2(1-p)$; check $p = 0$, 0.5 and 1; plot.	0 at $p = 0$ and 1; maximum 1 bit at $p = 0.5$

9. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Signals and spectrum			
1	21-22 1(b)	Diagram	ground, sky and line-of-sight propagation
2	21-22 1(d)	Graph	flat magnitude and linear phase of a distortionless channel
3	20-21 1(b)	Diagram	the three bands on a frequency scale
4	20-21 TT1 7(b)	Graph	flat magnitude and linear phase of a distortionless channel
5	19-20 1(a)	Diagram	twisted pair cancelling interference
6	19-20 1(a)	Diagram	ground, sky and line-of-sight propagation
Noise and channel capacity			
7	21-22 TT1 7(b)	Block diagram	amplifiers in cascade and the single equivalent noise resistor
8	20-21 1(c)	Graph	transistor noise against frequency
9	20-21 TT1 7(d)	Block diagram	amplifiers in cascade and the single equivalent noise resistor
10	19-20 1(d)	Block diagram	amplifiers in cascade and the single equivalent noise resistor

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No.	Paper	Kind	Diagram
Amplitude modulation			
11	22-23 TT2 7(c)	Circuit	ring modulator
12	22-23 TT2 7(c)	Waveform	message, switching wave and output
13	22-23 TT2 7(f)	Block diagram	Weaver SSB modulator
14	22-23 TT2 7(g)	Spectrum	VSB filter near the carrier
15	21-22 2(b)	Waveform	detected output with phase and frequency errors
16	21-22 2(c)	Block diagram	Costas receiver
17	21-22 2(d)	Block diagram	VSB transmitter and receiver
18	21-22 2(d)	Spectrum	VSB filter with complementary symmetry
19	21-22 3(b)	Circuit	envelope detector
20	21-22 3(b)	Waveform	AM input and detector output
21	21-22 3(d)	Waveform	AM for $\mu = 0.5$, 1 and 1.5
22	21-22 TT1 7(d)	Circuit	envelope detector
23	21-22 TT1 7(d)	Waveform	detector output, with ripple and diagonal clipping
24	21-22 TT2 8(a)	Block diagram	Costas receiver
25	21-22 TT2 8(c)	Spectrum	two-tone DSB-SC lines, the VSB filter and the detected lines
26	20-21 1(d)	Waveform	the AM wave
27	20-21 1(d)	Spectrum	carrier and the two side lines
28	20-21 4(b)	Block diagram	VSB transmitter and receiver
29	20-21 4(b)	Spectrum	VSB filter and the two shifted copies
30	20-21 5(d)	Spectrum	modulated and detected spectra for the two carrier frequencies
31	20-21 TT2 8(a)	Spectrum	DSB-WC and SSB-SC of the given message
32	19-20 1(b)	Spectrum	DSB-SC lines for the three messages
33	19-20 2(c)	Waveform	detected output with no error, a phase error and a frequency error
34	19-20 2(d)	Spectrum	speech spectrum and the SSB filter in the gap
35	19-20 3(a)	Block diagram	phase-shift SSB modulator
Angle modulation			
36	22-23 TT2 7(j)	Block diagram	NBFM generator
37	22-23 TT3 8(a)	Block diagram	Armstrong modulator with f_c and Δf at each stage
38	21-22 3(a)	Block diagram	NBFM generator
39	20-21 2(c)	Block diagram	FM from a phase modulator and PM from a frequency modulator
40	20-21 2(c)	Waveform	FM and PM for a triangular message
41	20-21 4(d)	Spectrum	FM lines for $\beta = 5$
42	20-21 TT2 8(d)	Circuit	pre-emphasis and de-emphasis networks
43	20-21 TT2 8(d)	Graph	their frequency responses
44	19-20 3(b)	Spectrum	FM lines for the reference, doubled amplitude and doubled frequency
45	19-20 3(c)	Block diagram	Armstrong modulator with f_c and Δf at each stage
46	19-20 4(b)	Block diagram	slope detector
47	19-20 4(b)	Graph	tuned-circuit response with the carrier on its slope

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No.	Paper	Kind	Diagram
Receivers			
48	20-21 4(a)	Block diagram	superheterodyne receiver
49	20-21 4(a)	Spectrum	the signal and its image frequency
Sampling and pulse modulation			
50	21-22 4(a)	Block diagram	PCM transmitter, channel and receiver
51	21-22 4(b)	Spectrum	sampled spectrum and the reconstruction LPF
52	21-22 4(b)	Waveform	sinc reconstruction
53	21-22 4(c)	Block diagram	compander chain
54	21-22 5(a)	Block diagram	DM and sigma-delta side by side
55	21-22 TT3 9(a)	Block diagram	ECG sampling, quantizing, coding and multiplexing
56	20-21 3(a)	Waveform	DM and ADM staircases following the same signal
57	20-21 3(b)	Block diagram	DM transmitter and receiver
58	20-21 3(c)	Block diagram	PCM system
59	20-21 3(c)	Waveform	sampling, quantizing and encoding
60	20-21 4(c)	Spectrum	sampled signal for three sampling rates
61	20-21 4(c)	Waveform	reconstruction by sinc pulses
62	19-20 4(c)	Graph	uniform and non-uniform quantizer characteristics
63	19-20 4(e)	Block diagram	DM transmitter and receiver
64	19-20 5(a)	Block diagram	DM and sigma-delta side by side
Digital modulation			
65	22-23 TT3 8(b)	Block diagram	DPSK modulator and demodulator
66	22-23 TT3 8(b)	Waveform	data, encoded bits, DPSK signal and receiver output
67	21-22 4(d)	Waveform	ASK, FSK and PSK for the same bits
68	21-22 5(b)	Block diagram	QAM modulator and coherent demodulator
69	21-22 5(c)	Table	DPSK values at each stage
70	21-22 5(d)	Block diagram	coherent and non-coherent BASK detectors
71	21-22 5(d)	Block diagram	coherent and non-coherent BFSK detectors
72	20-21 6(a)	Waveform	FSK for 10110
73	20-21 6(a)	Block diagram	two ways to generate FSK
74	20-21 6(a)	Spectrum	FSK
75	20-21 6(b)	Block diagram	QAM modulator and coherent demodulator
76	20-21 6(d)	Table	DPSK values at each stage
77	19-20 5(b)	Waveform	Manchester, AMI and dicode for 1010011100
78	19-20 5(c)	Block diagram	DPSK transmitter and receiver
79	19-20 5(d)	Block diagram	coherent BFSK detector
80	19-20 6(a)	Block diagram	QPSK modulator
81	19-20 6(a)	Constellation	the four QPSK points
Multiplexing and multiple access			
82	22-23 TT3 8(c)	Frame	pulse stuffing of the two channels
83	21-22 6(a)	Diagram	multilevel multiplexing
84	21-22 6(a)	Frame	multiple-slot TDM
85	21-22 6(b)	Block diagram	ECG sampling, quantizing, coding and multiplexing
86	21-22 6(e)	Diagram	multiplexing and multiple access
87	21-22 TT3 9(b)	Diagram	multilevel multiplexing

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No.	Paper	Kind	Diagram
88	21-22 TT3 9(b)	Frame	multiple-slot allocation
89	21-22 TT3 9(c)	Frame	four characters and a sync bit
90	20-21 2(a)	Diagram	multiplexing and multiple access
91	20-21 2(b)	Spectrum	five channels and four guard bands
92	20-21 5(a)	Diagram	TDMA and CDMA in time and frequency
93	20-21 5(c)	Block diagram	ECG sampling, quantizing, coding and multiplexing
94	20-21 6(c)	Block diagram	FHSS transmitter and receiver
95	20-21 6(c)	Graph	hopping pattern of eight frequencies
96	19-20 6(b)	Frame	TDM output of the four channels
97	19-20 6(c)	Block diagram	FDM multiplexer and demultiplexer
98	19-20 6(c)	Spectrum	channels side by side with guard bands
99	19-20 6(d)	Frame	pulse stuffing
100	19-20 6(d)	Frame	multiple-slot allocation
Information theory			
101	20-21 1(a)	Graph	entropy against probability
102	20-21 TT1 7(a)	Graph	entropy against probability

10. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Signals and spectrum		
Twisted pair vs radio waves	guided, noise cancelled by twisting vs unguided, omnidirectional, passes walls	19-20 1(a)
Radio waves vs microwaves vs infrared	omnidirectional, pass walls vs directional, line of sight vs short range, blocked by walls	20-21 1(b)
Amplitude vs phase vs nonlinear distortion	gain varies with frequency vs delay varies with frequency vs new frequencies appear	20-21 TT1 7(b)
Noise and channel capacity		
Thermal vs transit-time noise	at all frequencies, from heat vs only at very high frequency, rises with frequency	20-21 1(c)
Amplitude modulation		
Phase error vs frequency error	output scaled by $\cos\delta$, null at 90° vs output beats as $\cos\Delta\omega t$	22-23 TT2 7(e), 21-22 2(b), 20-21 TT2 8(b), 19-20 2(c)
SSB by filtering vs SSB by phase shift	needs a gap at dc and a sharp filter vs needs a wideband $-90°$ shifter	19-20 2(d)
AM vs DSB-SC vs SSB-SC	$2B$, envelope detector, at most 33 % vs $2B$, coherent, 100 % vs B, coherent, hardest modulator	22-23 TT2 7(d)
Weaver vs phase-shift SSB	two quadrature stages at fixed frequencies, no wideband shifter vs one wideband $-90°$ shifter	22-23 TT2 7(f)
SSB vs VSB vs DSB	bandwidth B, needs a gap at dc vs a little over B, keeps dc vs $2B$	22-23 TT2 7(g)
Angle modulation		
Message amplitude vs frequency in FM	sets Δf and the bandwidth vs sets the line spacing and β only	19-20 3(b)

Pair	Key difference	Asked in
Frequency multiplier vs mixer	multiplies both f_c and Δf vs moves f_c, keeps Δf	22-23 TT3 8(a), 19-20 3(c)
FM vs PM	frequency follows $m(t)$ vs phase follows $m(t)$	20-21 2(c)
NBFM vs AM	sideband term in quadrature with the carrier vs in phase with it; same bandwidth $2B$	21-22 3(a)
FM vs AM	noise immune, constant envelope, wide band vs narrow band, simple, longer range, noisy	22-23 TT2 7(a)
Sampling and pulse modulation		
Uniform vs non-uniform quantization	equal steps, SQNR falls for weak signals vs steps grow with amplitude, SQNR nearly constant	19-20 4(c)
DM vs sigma-delta	receiver accumulates, errors pile up, slope overload depends on frequency vs receiver is an LPF, no build-up	21-22 5(a), 19-20 5(a)
Slope overload vs granular noise	step too small for a steep signal vs step too large for a flat one	20-21 3(a)
Digital modulation		
Manchester vs AMI vs dicode	a transition in every bit vs marks alternate in sign vs pulses only at changes	19-20 5(b)
PSK vs DPSK	absolute phase, needs a coherent carrier vs phase change, no local carrier	19-20 5(c)
BPSK vs QPSK	one bit per symbol vs two bits per symbol, half the bandwidth, same error rate	19-20 6(a)
ASK vs FSK vs PSK	amplitude, worst in noise vs frequency, most bandwidth vs phase, best in noise	21-22 4(d)
Coherent vs non-coherent detection	needs a synchronized carrier, better in noise vs envelope only, simpler, worse in noise	21-22 5(d)
Multiplexing and multiple access		
Pulse stuffing vs multiple-slot allocation	rates not multiples, pad with dummy bits vs rates multiples of a base, give extra slots	19-20 6(d)
Multiplexing vs multiple access	signals combined at one point onto one link vs separate users share one medium	21-22 6(e), 20-21 2(a)
TDMA vs CDMA	separate time slots, hard capacity vs separate codes, soft capacity, power control	20-21 5(a)
Multilevel multiplexing vs multiple-slot allocation	slow inputs merged first into a faster stream vs a fast input gets several slots per frame	21-22 6(a), 21-22 TT3 9(b)
FDMA vs CDMA	own band with guard bands, hard capacity vs shared band separated by codes, soft capacity	21-22 6(c)

11. Waveform drills:

Every drawing drill asked, in book order: what to draw and the data given. The data changes from paper to paper; the method does not.

Draw	Data given	Asked in
Amplitude modulation		
DSB-WC and SSB-SC spectra	the triangular spectrum of Figure-1	20-21 TT2 8(a)
AM waveform for $\mu = 0.5$, 1, 1.5	the sawtooth $m(t)$ of Fig. 1	21-22 3(d)
Digital modulation		
Line codes (Manchester, AMI, dicode)	1010011100	19-20 5(b)

Draw	Data given	Asked in
DPSK encoder and receiver table	1010011100; 1010010110	22-23 TT3 8(b), 21-22 5(c), 20-21 6(d)
Multiplexing and multiple access		
TDM output with 2-bit slots	110010, 001010, 101101, 000111	19-20 6(b)

12. Repeated questions:

14 questions came back, exactly or with new numbers; 5 of them in three or more papers. Most repeated first; the latest paper first in each row.

No.	Times	Topic	Subtopic	Asked in (marks)
1	3	Digital modulation	DPSK	22-23 TT3 8(b) (4), 21-22 5(c) (6), 20-21 6(d) (4)
2	3	Noise and channel capacity	Cascaded noise	21-22 TT1 7(b) (5), 20-21 TT1 7(d) (5), 19-20 1(d) (7)
3	3	Amplitude modulation	DSB-SC	21-22 2(b) (6), 20-21 TT2 8(b) (5), 19-20 2(c) (6)
4	3	Multiplexing and multiple access	TDM	21-22 6(b) (6), 21-22 TT3 9(a) (7), 20-21 5(c) (7)
5	3	Amplitude modulation	AM modulation index	21-22 2(a) (5), 21-22 TT1 7(c) (5), 20-21 TT2 8(c) (5)
6	2	Angle modulation	Armstrong modulator	22-23 TT3 8(a) (10), 19-20 3(c) (9)
7	2	Amplitude modulation	Sidebands and power	22-23 TT2 7(b), 20-21 1(e) (6)
8	2	Sampling and pulse modulation	Delta modulation	21-22 5(a) (6), 19-20 5(a) (5)
9	2	Multiplexing and multiple access	Multiple access	21-22 6(e) (3), 20-21 2(a) (5)
10	2	Digital modulation	QAM	21-22 5(b) (7), 20-21 6(b) (7)
11	2	Signals and spectrum	Distortionless transmission	21-22 1(d) (6), 20-21 TT1 7(b) (5)
12	2	Noise and channel capacity	Channel capacity	21-22 1(c) (6), 20-21 TT1 7(c)
13	2	Multiplexing and multiple access	TDM	21-22 6(a) (6), 21-22 TT3 9(b) (6)
14	2	Information theory	Entropy	20-21 1(a) (5), 20-21 TT1 7(a) (5)

A sub-part with no marks after it had none printed on its paper.

13. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
Signals and spectrum	3	2	·	·	·
Noise and channel capacity	1	3	5	·	·
Amplitude modulation	11	9	7	2	·
Angle modulation	7	1	5	1	2
Receivers	1	·	·	·	·
Sampling and pulse modulation	7	2	3	2	·
Digital modulation	8	·	2	2	·
Multiplexing and multiple access	9	·	6	·	·

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Topic	Theory	Derivation	Numerical	Diagram	Design
Information theory	·	1	3	1	·
System design	1	·	·	·	·

14. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Amplitude modulation	125	22 %
2	Angle modulation	85	37 %
3	Multiplexing and multiple access	82	52 %
4	Digital modulation	75	65 %
5	Sampling and pulse modulation	74	78 %
6	Noise and channel capacity	49	87 %
7	Signals and spectrum	30	92 %
8	Information theory	29	98 %
9	Receivers	7	99 %
10	System design	7	100 %

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